Weakly Nonlinear Reduction with Separated Temporal and Spatial Forcing

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1 Problem setting

We consider

$$\begin{aligned} u_t = {} & \mathcal{L}(p)u + \mathcal{Q}(u,u) + \mathcal{C}(u,u,u) \\ & + \epsilon^n \left(P\mathrm{e}^{\mathrm{i}\omega t} + \overline{P}\mathrm{e}^{-\mathrm{i}\omega t} \right) + \epsilon^m \left(S\mathrm{e}^{\mathrm{i}kx} + \overline{S}\mathrm{e}^{-\mathrm{i}kx} \right), \end{aligned} \tag{1}$$

where $u(x,t) \in \mathbb{R}^d$, $x \in \mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$, $p \in \mathbb{R}$, and

$$\mathcal{L}(p) = \mathcal{L}_0 + p\mathcal{L}_1 + O(p^2). \tag{2}$$

The forcing vectors are denoted by P, S so that $\mathcal{Q}$ is reserved for the quadratic map.

Assume:

1. $\mathcal{Q}$ and $\mathcal{C}$ are symmetric multilinear maps, extended complex linearly;
2. $\mathcal{L}_j$ is real, translation invariant, and reflection invariant, $j = 0, 1$;
3. the only critical modes are the simple finite-wavenumber Hopf modes $(\Omega, h) = (\pm\omega_0, \pm k_0)$, where $\omega_0 > 0$ and $k_0 \in \mathbb{N}$.

Here reflection invariance is the intended symmetry of $\mathcal{L}_j$; it does not mean that $\mathcal{L}_j$ is self-adjoint.

For a spatial Fourier mode, define

$$\mathcal{L}_j \left(v\mathrm{e}^{\mathrm{i}hx} \right) = M_j(h) v \mathrm{e}^{\mathrm{i}hx}, \qquad h \in \mathbb{Z}. \tag{3}$$

Reality and reflection symmetry give, respectively,

$$M_j(-h) = \overline{M_j(h)}, \qquad M_j(-h) = M_j(h). \tag{4}$$

Thus $M_j(h)$ is real in this setting and the matrices at h and $-h$ are identical.

Choose right and adjoint eigenvectors $\Phi, \Psi \in \mathbb{C}^d$ such that

$$M_0(\pm k_0)\Phi = \mathrm{i}\omega_0\Phi, \qquad M_0(\pm k_0)^*\Psi = -\mathrm{i}\omega_0\Psi, \qquad \Psi^*\Phi = 1. \tag{5}$$

The conjugate equations give the eigenvectors $\overline{\Phi}$ and $\overline{\Psi}$ at eigenvalues $-\mathrm{i}\omega_0$ and $+\mathrm{i}\omega_0$ of M_0 and M_0^*, respectively. Because of (4), the same Φ, Ψ can be used at $\pm k_0$.

Let $\nu(h,p)$ be the eigenvalue branch through $\nu(\pm k_0,0)=\mathrm{i}\omega_0$. If $M(\pm k_0,p)\Phi(p)=\nu(\pm k_0,p)\Phi(p)$, differentiation at $p=0$ gives

$$M_1(\pm k_0)\Phi + M_0(\pm k_0)\Phi_p = \nu_p\Phi + \mathrm{i}\omega_0\Phi_p. \tag{6}$$

Since $\Psi^* M_0(\pm k_0) = \mathrm{i}\omega_0\Psi^*$, multiplication by Ψ^* cancels both terms containing Φ_p. Therefore

$$\partial_p\nu(\pm k_0,0) = \Psi^* M_1(\pm k_0)\Phi =: \alpha, \qquad \operatorname{Re}\alpha > 0. \tag{7}$$

The last inequality is the transverse-crossing assumption. In particular, $\alpha \neq 0$.

Define the modal resolvent

$$R(\Omega,h) := (\mathrm{i}\Omega I - M_0(h))^{-1}. \tag{8}$$

The no-quadratic-resonance assumption is

$$\begin{matrix} 2\mathrm{i}\omega_0 I - M_0(0), & 2\mathrm{i}\omega_0 I - M_0(\pm 2k_0), \\ -M_0(0), & -M_0(\pm 2k_0) \end{matrix} \quad \text{are invertible.} \tag{9}$$

We additionally assume that $R(\pm\omega,0)$ and $R(0,\pm k)$ exist whenever the corresponding forcing is present. Failure of any of these invertibilities introduces an additional center mode and requires a larger amplitude system.

Introduce fast and slow times

$$s = t, \qquad \tau = \epsilon^2 t, \qquad \partial_t = \partial_s + \epsilon^2\partial_\tau, \tag{10}$$

and initially write

$$p = \epsilon p_1 + \epsilon^2 p_2 + O(\epsilon^3), \qquad u = \epsilon u_1 + \epsilon^2 u_2 + \epsilon^3 u_3 + O(\epsilon^4). \tag{11}$$

The terms that determine the first three equations are

$$\partial_t u = \epsilon\partial_s u_1 + \epsilon^2\partial_s u_2 + \epsilon^3(\partial_s u_3 + \partial_\tau u_1) + O(\epsilon^4), \tag{12}$$
$$\mathcal{L}_0 u = \epsilon\mathcal{L}_0 u_1 + \epsilon^2\mathcal{L}_0 u_2 + \epsilon^3\mathcal{L}_0 u_3 + O(\epsilon^4), \tag{13}$$
$$p\mathcal{L}_1 u = \epsilon^2 p_1\mathcal{L}_1 u_1 + \epsilon^3(p_2\mathcal{L}_1 u_1 + p_1\mathcal{L}_1 u_2) + O(\epsilon^4), \tag{14}$$
$$\mathcal{Q}(u,u) = \epsilon^2\mathcal{Q}(u_1,u_1) + 2\epsilon^3\mathcal{Q}(u_1,u_2) + O(\epsilon^4), \tag{15}$$
$$\mathcal{C}(u,u,u) = \epsilon^3\mathcal{C}(u_1,u_1,u_1) + O(\epsilon^4). \tag{16}$$

The factor 2 in (15) is the symmetry of $\mathcal{Q}$. The term $p_1\mathcal{L}_1 u_2$ and every higher-p contribution at $O(\epsilon^3)$ vanish after the second-order solvability condition gives $p_1 = 0$.

2 The solution space of $\mathcal{D} = \partial_s - \mathcal{L}_0$

Set

$$\mathcal{D} := \partial_s - \mathcal{L}_0, \qquad E_\pm := \mathrm{e}^{\mathrm{i}(\omega_0 s \pm k_0 x)}. \tag{17}$$

For any pure wave,

$$\mathcal{D}\left(v\mathrm{e}^{\mathrm{i}(\Omega s + hx)}\right) = (\mathrm{i}\Omega I - M_0(h))\, v\mathrm{e}^{\mathrm{i}(\Omega s + hx)}. \tag{18}$$

The multiplier in (18) is singular at $(\Omega, h) = (\omega_0, \pm k_0)$ with nullvector Φ, and its conjugate is singular at $(-\omega_0, \mp k_0)$ with nullvector $\overline{\Phi}$. There are no other singular Fourier blocks by assumption. Hence

$$\ker \mathcal{D} = \operatorname{span}_{\mathbb{R}} \left\{ \Phi E_+, \Phi E_-, \overline{\Phi}\,\overline{E}_+, \overline{\Phi}\,\overline{E}_- \right\}. \tag{19}$$

On functions periodic in x and in s with $T_0 = 2\pi/\omega_0$, use

$$\langle v, w \rangle := \frac{1}{2\pi T_0} \int_0^{T_0} \int_0^{2\pi} v(s,x)^* w(s,x)\, dx\, ds. \tag{20}$$

Periodic integration by parts gives

$$\mathcal{D}^* = -\partial_s - \mathcal{L}_0^*. \tag{21}$$

Indeed,

$$\mathcal{D}^* \left(w e^{\mathrm{i}(\Omega s + hx)} \right) = \left(-\mathrm{i}\Omega I - M_0(h)^* \right) w e^{\mathrm{i}(\Omega s + hx)}, \tag{22}$$

so (5) gives

$$\ker \mathcal{D}^* = \operatorname{span}_{\mathbb{R}} \left\{ \Psi E_+, \Psi E_-, \overline{\Psi}\,\overline{E}_+, \overline{\Psi}\,\overline{E}_- \right\}. \tag{23}$$

For pure waves,

$$\left\langle a e^{\mathrm{i}(\Omega s + hx)}, b e^{\mathrm{i}(\Omega' s + h' x)} \right\rangle = a^* b\, \delta_{\Omega,\Omega'} \delta_{h,h'}. \tag{24}$$

Thus different wave types are orthogonal. Having the same frequency–wavenumber pair is necessary, but not sufficient, for a nonzero inner product because a^*b may also vanish. For incommensurate forcing frequencies, (24) is interpreted as a long-time mean; equivalently, one works on the torus of independent fast phases.

Let

$$U_\pm = \Phi E_\pm, \qquad V_\pm = \Psi E_\pm. \tag{25}$$

The complete pairings needed below are

	$\mathcal{L}_1 U_+$	$\mathcal{L}_1 U_-$	$\mathcal{L}_1 \overline{U}_+$	$\mathcal{L}_1 \overline{U}_-$
V_+	α	0	0	0
V_-	0	α	0	0
$\overline{V}_+$	0	0	$\overline{\alpha}$	0
$\overline{V}_-$	0	0	0	$\overline{\alpha}$

. (26)

Under the spectral assumptions above, $\mathcal{D}$ is Fredholm with finite-dimensional kernel and closed range: in Fourier variables it is the direct sum of the blocks in (18), all noncritical blocks are invertible, and the critical part is finite dimensional. Hence

$$\operatorname{Ran} \mathcal{D} = (\ker \mathcal{D}^*)^\perp, \qquad \mathcal{D}w = f \iff \langle V, f \rangle = 0 \quad \text{for every } V \in \ker \mathcal{D}^*. \tag{27}$$

At every order, the goal is to put the right-hand side in $\operatorname{Ran} \mathcal{D}$.

To prevent the stimulus itself from supplying the leading pattern, take $n, m > 1$. If $n = 1$ or $m = 1$, the corresponding stimulus appears in $\mathcal{D}u_1$ and u_1 is no longer chosen solely from $\ker \mathcal{D}$. If an exponent equals 2, its response is part of u_2 and interacts with u_1 through $2\mathcal{Q}(u_1, u_2)$ at order three. An exponent greater than 2 cannot produce such a cubic-order feedback. In particular, exponent-3 forcing would enter (29c) directly, but its separated types $(\omega, 0)$ and $(0, \pm k)$ are never critical because $k_0, \omega_0 > 0$; it changes only u_3. Exponents larger than 3 do not occur in the present

truncation. Thus only exponent-2 forcing can modify the leading amplitude equations, and we define

$$F_2 = \mathbf{1}_{\{n=2\}} \left(P\mathrm{e}^{\mathrm{i}\omega s} + \overline{P}\mathrm{e}^{-\mathrm{i}\omega s}\right) + \mathbf{1}_{\{m=2\}} \left(S\mathrm{e}^{\mathrm{i}kx} + \overline{S}\mathrm{e}^{-\mathrm{i}kx}\right). \tag{28}$$

The three equations are

$$O(\epsilon): \quad \mathcal{D}u_1 = 0, \tag{29a}$$

$$O(\epsilon^2): \quad \mathcal{D}u_2 = p_1\mathcal{L}_1u_1 + \mathcal{Q}(u_1,u_1) + F_2, \tag{29b}$$

$$O(\epsilon^3): \quad \mathcal{D}u_3 = -\partial_\tau u_1 + p_2\mathcal{L}_1u_1 + 2\mathcal{Q}(u_1,u_2) + \mathcal{C}(u_1,u_1,u_1), \tag{29c}$$

where (29c) is written after the second-order condition below sets $p_1 = 0$. We subsequently write $p_2 = \mu$.

3 $O(\epsilon)$ solution

The real leading solution is

$$\boxed{u_1 = A(\tau)\Phi E_+ + B(\tau)\Phi E_- + \text{c.c.}.} \tag{30}$$

The amplitudes A and B are independent. The states $B = 0$ and $A = 0$ are the two traveling-wave directions. Equal moduli $|A| = |B|$ give standing waves; imposing $A = B$ at the outset would restrict the calculation to one standing-wave subspace.

4 $O(\epsilon^2)$ solution

The right-hand side of (29b) must be orthogonal to $\ker\mathcal{D}^*$. Adding any two wave types from $(\pm\omega_0, \pm k_0)$ produces one of the types in (40) below, never $(\omega_0, \pm k_0)$. Hence

$$\langle V_\pm, \mathcal{Q}(u_1,u_1)\rangle = 0. \tag{31}$$

Neither separated forcing is a critical wave because $(\omega, 0) \neq (\omega_0, \pm k_0)$ and $(0, \pm k) \neq (\omega_0, \pm k_0)$, so

$$\langle V_\pm, F_2\rangle = 0. \tag{32}$$

Only $p_1\mathcal{L}_1u_1$ remains. By (26),

$$\langle V_+, p_1\mathcal{L}_1u_1\rangle = p_1A\alpha, \qquad \langle V_-, p_1\mathcal{L}_1u_1\rangle = p_1B\alpha. \tag{33}$$

Thus the two independent complex conditions are

$$0 = p_1A\alpha, \qquad 0 = p_1B\alpha. \tag{34}$$

For $(A, B) \neq (0, 0)$ and $\operatorname{Re}\alpha > 0$,

$$\boxed{p_1 = 0, \qquad p = \epsilon^2\mu + O(\epsilon^3).} \tag{35}$$

Define

$$q_{20} := \mathcal{Q}(\Phi,\Phi), \qquad q_{11} := \mathcal{Q}(\Phi,\overline{\Phi}). \tag{36}$$

To track the factors of two, set

$$u_1^+ := \Phi(AE_+ + BE_-), \qquad u_1 = u_1^+ + \overline{u_1^+}. \tag{37}$$

Symmetry of $\mathcal{Q}$ gives

$$\mathcal{Q}(u_1, u_1) = \mathcal{Q}(u_1^+, u_1^+) + 2\mathcal{Q}(u_1^+, \overline{u_1^+}) + \mathcal{Q}(\overline{u_1^+}, \overline{u_1^+}), \tag{38}$$

and therefore

$$\begin{aligned}\mathcal{Q}(u_1, u_1) = {} & q_{20}\left(A^2E_+^2 + B^2E_-^2 + 2ABE_+E_-\right) + \text{c.c.} \\ & + 2q_{11}\left(|A|^2 + |B|^2 + A\overline{B}E_+\overline{E}_- + \overline{A}B\overline{E}_+E_-\right).\end{aligned} \tag{39}$$

The generated wave types are

$$(\pm 2\omega_0, \pm 2k_0), \quad (\pm 2\omega_0, 0), \quad (0, \pm 2k_0), \quad (0, 0). \tag{40}$$

They are noncritical by wave-type arithmetic; condition (9) is the additional statement that $\mathcal{D}$ is invertible on each of them.

For a noncritical type, (18) gives the mode-by-mode rule

$$\mathcal{D}^{-1}\left(f\mathrm{e}^{\mathrm{i}(\Omega s + hx)}\right) = R(\Omega, h) f \mathrm{e}^{\mathrm{i}(\Omega s + hx)}. \tag{41}$$

Set

$$h_{20}^+ = R(2\omega_0, 2k_0)q_{20}, \qquad h_{20}^- = R(2\omega_0, -2k_0)q_{20}, \tag{42a}$$

$$h_{20}^0 = 2R(2\omega_0, 0)q_{20}, \qquad h_{11}^0 = 2R(0, 0)q_{11}, \tag{42b}$$

$$h_{11}^+ = 2R(0, 2k_0)q_{11}, \qquad h_{11}^- = 2R(0, -2k_0)q_{11}. \tag{42c}$$

The six nonconjugate source-to-response steps are

source in $\mathcal{Q}(u_1, u_1)$	(Ω, h)	term in u_2^Q
$A^2q_{20}E_+^2$	$(2\omega_0, 2k_0)$	$A^2h_{20}^+E_+^2$
$B^2q_{20}E_-^2$	$(2\omega_0, -2k_0)$	$B^2h_{20}^-E_-^2$
$2ABq_{20}E_+E_-$	$(2\omega_0, 0)$	$ABh_{20}^0E_+E_-$
$2(\lvert A\rvert^2 + \lvert B\rvert^2)q_{11}$	$(0, 0)$	$(\lvert A\rvert^2 + \lvert B\rvert^2)h_{11}^0$
$2A\overline{B}q_{11}E_+\overline{E}_-$	$(0, 2k_0)$	$A\overline{B}h_{11}^+E_+\overline{E}_-$
$2\overline{A}Bq_{11}\overline{E}_+E_-$	$(0, -2k_0)$	$\overline{A}Bh_{11}^-\overline{E}_+E_-$

. (43)

Then

$$\begin{aligned}u_2^Q := {} & \mathcal{D}^{-1}\mathcal{Q}(u_1, u_1) \\ = {} & A^2h_{20}^+E_+^2 + B^2h_{20}^-E_-^2 + ABh_{20}^0E_+E_- + \text{c.c.} \\ & + (|A|^2 + |B|^2)h_{11}^0 + A\overline{B}h_{11}^+E_+\overline{E}_- + \overline{A}Bh_{11}^-\overline{E}_+E_-.\end{aligned} \tag{44}$$

Reflection symmetry gives $h_{20}^+ = h_{20}^-$ and $h_{11}^+ = h_{11}^-$. The superscripts are retained to display the wave types.

For the general forcing frequency and wavenumber, define

$$p_\omega^+ = R(\omega, 0)P, \qquad p_\omega^- = R(-\omega, 0)\overline{P}, \tag{45a}$$

$$s_k^+ = R(0, k)S, \qquad s_k^- = R(0, -k)\overline{S}. \tag{45b}$$

For example,

$$\mathcal{D}\left(p_\omega^+ \mathrm{e}^{\mathrm{i}\omega s}\right) = (\mathrm{i}\omega I - M_0(0))p_\omega^+ \mathrm{e}^{\mathrm{i}\omega s} = P\mathrm{e}^{\mathrm{i}\omega s}, \tag{46}$$

and the other three forcing modes are identical applications of (41). Thus

$$\begin{aligned} u_2^F &= \mathcal{D}^{-1}F_2 \\ &= \mathbf{1}_{\{n=2\}}\left(p_\omega^+ \mathrm{e}^{\mathrm{i}\omega s} + p_\omega^- \mathrm{e}^{-\mathrm{i}\omega s}\right) + \mathbf{1}_{\{m=2\}}\left(s_k^+ \mathrm{e}^{\mathrm{i}kx} + s_k^- \mathrm{e}^{-\mathrm{i}kx}\right). \end{aligned} \tag{47}$$

Reality gives $p_\omega^- = \overline{p_\omega^+}$ and $s_k^- = \overline{s_k^+}$. We use the normalization

$$\boxed{u_2 = u_2^Q + u_2^F,} \tag{48}$$

with no added component from $\ker \mathcal{D}$.

5 $O(\epsilon^3)$ solution and solvability

With $p_2 = \mu$, write the right-hand side of (29c) in four groups:

$$F_{0,\mathrm{lin}}^{(3)} := -\partial_\tau u_1 + \mu\mathcal{L}_1 u_1, \tag{49}$$

$$F_{0,\mathrm{int}}^{(3)} := 2\mathcal{Q}(u_1, u_2^Q) + \mathcal{C}(u_1, u_1, u_1), \tag{50}$$

$$F_{F,\mathrm{lin}}^{(3)} := 0, \tag{51}$$

$$F_{F,\mathrm{int}}^{(3)} := 2\mathcal{Q}(u_1, u_2^F). \tag{52}$$

The third line is zero because both retained stimuli entered at order two; their only cubic-order effect is through u_2^F.

5.1 Unforced terms

For the critical wave E_+,

$$[F_{0,\mathrm{lin}}^{(3)}]_{E_+} = -A_\tau \Phi + \mu A M_1(k_0)\Phi. \tag{53}$$

The resonant cubic contribution is

$$[\mathcal{C}(u_1, u_1, u_1)]_{E_+} = 3A|A|^2\mathcal{C}(\Phi, \Phi, \overline{\Phi}) + 6A|B|^2\mathcal{C}(\Phi, \Phi, \overline{\Phi}). \tag{54}$$

Indeed, a triple of critical phases can sum to E_+ only as $E_+E_+\overline{E}_+$ or $E_+E_-\overline{E}_-$. Their permutation counts are 3 and 6, respectively, which gives (54).

For the quadratic feedback, frequency and wavenumber must add separately. The complete E_+ selection table is

factor from u_1	factor from u_2^Q	coefficient after $2\mathcal{Q}$
$A\Phi E_+$	$(\lvert A\rvert^2 + \lvert B\rvert^2)h_{11}^0$	$2A(\lvert A\rvert^2 + \lvert B\rvert^2)\mathcal{Q}(\Phi, h_{11}^0)$
$\overline{A}\,\overline{\Phi}\,\overline{E}_+$	$A^2 h_{20}^+ E_+^2$	$2A\lvert A\rvert^2\mathcal{Q}(\overline{\Phi}, h_{20}^+)$
$\overline{B}\,\overline{\Phi}\,\overline{E}_-$	$ABh_{20}^0 E_+E_-$	$2A\lvert B\rvert^2\mathcal{Q}(\overline{\Phi}, h_{20}^0)$
$B\Phi E_-$	$A\overline{B}h_{11}^+ E_+\overline{E}_-$	$2A\lvert B\rvert^2\mathcal{Q}(\Phi, h_{11}^+)$

. (55)

No other pair of wave types sums to (ω_0, k_0). Summing the last column gives

$$\begin{aligned}[2\mathcal{Q}(u_1, u_2^Q)]_{E_+} = {} & 2A(|A|^2 + |B|^2)\mathcal{Q}(\Phi, h_{11}^0) \\ & + 2A|A|^2\mathcal{Q}(\overline{\Phi}, h_{20}^+) + 2A|B|^2\mathcal{Q}(\overline{\Phi}, h_{20}^0) \\ & + 2A|B|^2\mathcal{Q}(\Phi, h_{11}^+). \end{aligned} \tag{56}$$

Define

$$G_{\mathrm{s}} = 2\mathcal{Q}(\Phi, h_{11}^0) + 2\mathcal{Q}(\overline{\Phi}, h_{20}^+) + 3\mathcal{C}(\Phi, \Phi, \overline{\Phi}), \tag{57a}$$

$$G_{\mathrm{c}} = 2\mathcal{Q}(\Phi, h_{11}^0) + 2\mathcal{Q}(\overline{\Phi}, h_{20}^0) + 2\mathcal{Q}(\Phi, h_{11}^+) + 6\mathcal{C}(\Phi, \Phi, \overline{\Phi}). \tag{57b}$$

Therefore

$$[F_{0,\mathrm{lin}}^{(3)} + F_{0,\mathrm{int}}^{(3)}]_{E_+} = -A_\tau\Phi + \mu A M_1(k_0)\Phi + A|A|^2 G_{\mathrm{s}} + A|B|^2 G_{\mathrm{c}}, \tag{58a}$$

$$[F_{0,\mathrm{lin}}^{(3)} + F_{0,\mathrm{int}}^{(3)}]_{E_-} = -B_\tau\Phi + \mu B M_1(k_0)\Phi + B|B|^2 G_{\mathrm{s}} + B|A|^2 G_{\mathrm{c}}. \tag{58b}$$

These are the only unforced terms outside $\operatorname{Ran}\mathcal{D}$.

5.2 Forcing-interaction terms and their wave types

Let $a_+ = A$, $a_- = B$, and let $\sigma \in \{+1, -1\}$. The complete temporal forcing interaction is

$$\begin{aligned}2\mathcal{Q}(u_1, u_{2,t}^F) = 2\mathbf{1}_{\{n=2\}} \sum_{\sigma=\pm1} \Big[& a_\sigma \mathcal{Q}(\Phi, p_\omega^+)\mathrm{e}^{\mathrm{i}((\omega_0+\omega)s+\sigma k_0 x)} \\ & + a_\sigma \mathcal{Q}(\Phi, p_\omega^-)\mathrm{e}^{\mathrm{i}((\omega_0-\omega)s+\sigma k_0 x)} \\ & + \overline{a_\sigma}\mathcal{Q}(\overline{\Phi}, p_\omega^+)\mathrm{e}^{\mathrm{i}((\omega-\omega_0)s-\sigma k_0 x)} \\ & + \overline{a_\sigma}\mathcal{Q}(\overline{\Phi}, p_\omega^-)\mathrm{e}^{-\mathrm{i}((\omega+\omega_0)s+\sigma k_0 x)}\Big]. \end{aligned} \tag{59}$$

Temporal forcing does not change spatial wavenumber. Therefore an E_+ term can only use the two first-order factors whose wavenumber is $+k_0$:

first-order type	matching equation	solution
(ω_0, k_0)	$(\omega_0, k_0) + (\pm\omega, 0) = (\omega_0, k_0)$	$\omega = 0$
$(-\omega_0, k_0)$	$(-\omega_0, k_0) + (\omega, 0) = (\omega_0, k_0)$	$\omega = 2\omega_0$

. (60)

The E_- calculation is its reflection. For $\omega \geq 0$, its positive-frequency critical pieces are

$$[2\mathcal{Q}(u_1, u_{2,t}^F)]_{E_+} = \mathbf{1}_{\{n=2\}}\left[\mathbf{1}_{\{\omega=0\}} 2A\mathcal{Q}(\Phi, p_0^+ + p_0^-) + \mathbf{1}_{\{\omega=2\omega_0\}} 2\overline{B}\mathcal{Q}(\overline{\Phi}, p_{2\omega_0}^+)\right], \tag{61}$$

$$[2\mathcal{Q}(u_1, u_{2,t}^F)]_{E_-} = \mathbf{1}_{\{n=2\}}\left[\mathbf{1}_{\{\omega=0\}} 2B\mathcal{Q}(\Phi, p_0^+ + p_0^-) + \mathbf{1}_{\{\omega=2\omega_0\}} 2\overline{A}\mathcal{Q}(\overline{\Phi}, p_{2\omega_0}^+)\right]. \tag{62}$$

The forcing terms that survive Fredholm projection therefore have fast wave types

$$\boxed{(2\omega_0, 0) + (-\omega_0, \pm k_0) = (\omega_0, \pm k_0).} \tag{63}$$

The complete spatial forcing interaction is

$$\begin{aligned}2\mathcal{Q}(u_1, u_{2,x}^F) = 2\mathbf{1}_{\{m=2\}} \sum_{\sigma=\pm1} \Big[& a_\sigma \mathcal{Q}(\Phi, s_k^+)\mathrm{e}^{\mathrm{i}(\omega_0 s+(\sigma k_0+k)x)} \\ & + a_\sigma \mathcal{Q}(\Phi, s_k^-)\mathrm{e}^{\mathrm{i}(\omega_0 s+(\sigma k_0-k)x)} \\ & + \overline{a_\sigma}\mathcal{Q}(\overline{\Phi}, s_k^+)\mathrm{e}^{\mathrm{i}(-\omega_0 s+(-\sigma k_0+k)x)} \\ & + \overline{a_\sigma}\mathcal{Q}(\overline{\Phi}, s_k^-)\mathrm{e}^{-\mathrm{i}(\omega_0 s+(\sigma k_0+k)x)}\Big]. \end{aligned} \tag{64}$$

Spatial forcing does not change temporal frequency. Hence an E_+ term must use a positive-frequency first-order factor, and the possible matches are

first-order type	matching equation	solution
(ω_0, k_0)	$(\omega_0, k_0) + (0, \pm k) = (\omega_0, k_0)$	$k = 0$
$(\omega_0, -k_0)$	$(\omega_0, -k_0) + (0, k) = (\omega_0, k_0)$	$k = 2k_0$

. (65)

Again, reflection gives the E_- calculation. For $k \geq 0$, its positive-frequency critical pieces are

$$[2\mathcal{Q}(u_1, u^F_{2,x})]_{E_+} = \mathbf{1}_{\{m=2\}} \left[\mathbf{1}_{\{k=0\}} 2A\mathcal{Q}(\Phi, s_0^+ + s_0^-) + \mathbf{1}_{\{k=2k_0\}} 2B\mathcal{Q}(\Phi, s^+_{2k_0}) \right], \tag{66}$$

$$[2\mathcal{Q}(u_1, u^F_{2,x})]_{E_-} = \mathbf{1}_{\{m=2\}} \left[\mathbf{1}_{\{k=0\}} 2B\mathcal{Q}(\Phi, s_0^+ + s_0^-) + \mathbf{1}_{\{k=2k_0\}} 2A\mathcal{Q}(\Phi, s^-_{2k_0}) \right]. \tag{67}$$

The nontrivial spatial match is

$$\boxed{(0, \pm 2k_0) + (\omega_0, \mp k_0) = (\omega_0, \pm k_0).} \tag{68}$$

Consequently, relative to the unforced problem, the complete list of cubic-order exceptions for $n, m > 1$ is

forcing	condition	critical term
temporal DC	$n = 2,\ \omega = 0$	$A,\ B$
temporal parametric	$n = 2,\ \omega = 2\omega_0$	$\overline{B},\ \overline{A}$
spatial DC	$m = 2,\ k = 0$	$A,\ B$
spatial parametric	$m = 2,\ k = 2k_0$	$B,\ A$

. (69)

If ω and k are restricted to be strictly positive, the two DC rows are absent. For all other ω, k, every term in (59) and (64) lies in $\operatorname{Ran} \mathcal{D}$ and only changes u_3.

5.3 Fredholm conditions

Define

$$a := \mu \Psi^* M_1(k_0) \Phi = \mu\alpha, \qquad b := \Psi^* G_{\mathrm{s}}, \qquad c := \Psi^* G_{\mathrm{c}}. \tag{70}$$

Because of the normalized fast phases and $\Psi^* \Phi = 1$,

$$\left\langle V_+, F^{(3)} \right\rangle = \Psi^* [F^{(3)}]_{E_+}, \qquad \left\langle V_-, F^{(3)} \right\rangle = \Psi^* [F^{(3)}]_{E_-}. \tag{71}$$

For completeness, the forcing projections are

$$\mathcal{T}_A = \mathbf{1}_{\{n=2\}} \left[\mathbf{1}_{\{\omega=0\}} 2A\Psi^* \mathcal{Q}(\Phi, p_0^+ + p_0^-) + \mathbf{1}_{\{\omega=2\omega_0\}} 2\overline{B}\Psi^* \mathcal{Q}(\overline{\Phi}, p^+_{2\omega_0}) \right], \tag{72a}$$

$$\mathcal{T}_B = \mathbf{1}_{\{n=2\}} \left[\mathbf{1}_{\{\omega=0\}} 2B\Psi^* \mathcal{Q}(\Phi, p_0^+ + p_0^-) + \mathbf{1}_{\{\omega=2\omega_0\}} 2\overline{A}\Psi^* \mathcal{Q}(\overline{\Phi}, p^+_{2\omega_0}) \right], \tag{72b}$$

$$\mathcal{X}_A = \mathbf{1}_{\{m=2\}} \left[\mathbf{1}_{\{k=0\}} 2A\Psi^* \mathcal{Q}(\Phi, s_0^+ + s_0^-) + \mathbf{1}_{\{k=2k_0\}} 2B\Psi^* \mathcal{Q}(\Phi, s^+_{2k_0}) \right], \tag{72c}$$

$$\mathcal{X}_B = \mathbf{1}_{\{m=2\}} \left[\mathbf{1}_{\{k=0\}} 2B\Psi^* \mathcal{Q}(\Phi, s_0^+ + s_0^-) + \mathbf{1}_{\{k=2k_0\}} 2A\Psi^* \mathcal{Q}(\Phi, s^-_{2k_0}) \right]. \tag{72d}$$

Projection onto V_+ and V_- now gives

$$0 = -A_\tau + A(a + b|A|^2 + c|B|^2) + \mathcal{T}_A + \mathcal{X}_A, \tag{73a}$$

$$0 = -B_\tau + B(a + b|B|^2 + c|A|^2) + \mathcal{T}_B + \mathcal{X}_B, \tag{73b}$$

The conjugate adjoint modes give only the conjugates of (73); hence these are the complete independent conditions.

6 Amplitude equations and pattern selection

6.1 No leading forcing resonance

When the forcing projections vanish, the amplitude equations are

$$A_\tau = A(a + b|A|^2 + c|B|^2), \tag{74a}$$
$$B_\tau = B(a + b|B|^2 + c|A|^2). \tag{74b}$$

Write $a_r = \operatorname{Re} a$, $b_r = \operatorname{Re} b$, and $c_r = \operatorname{Re} c$. For $A = r_+ e^{i\theta_+}$ and $B = r_- e^{i\theta_-}$,

$$(r_+)_\tau = r_+(a_r + b_r r_+^2 + c_r r_-^2), \tag{75a}$$
$$(r_-)_\tau = r_-(a_r + b_r r_-^2 + c_r r_+^2). \tag{75b}$$

The zero state has the double radial eigenvalue a_r and is stable for $a_r < 0$. For $a_r > 0$, a traveling wave satisfies

$$r_- = 0, \qquad a_r + b_r r_+^2 = 0, \qquad \rho^2 := r_+^2 = -\frac{a_r}{b_r}. \tag{76}$$

Its radial and transverse eigenvalues are

$$\lambda_{\mathrm{rad}} = -2a_r, \qquad \lambda_{\mathrm{tr}} = a_r + c_r\rho^2 = a_r\frac{b_r - c_r}{b_r}. \tag{77}$$

A standing wave satisfies

$$r_+ = r_- = \rho, \qquad \rho^2 = -\frac{a_r}{b_r + c_r}. \tag{78}$$

The radial Jacobian is

$$2\rho^2 \begin{pmatrix} b_r & c_r \\ c_r & b_r \end{pmatrix}, \tag{79}$$

with symmetric and antisymmetric eigenvalues

$$\lambda_{\mathrm{sym}} = -2a_r, \qquad \lambda_{\mathrm{asym}} = 2\rho^2(b_r - c_r). \tag{80}$$

Consequently the nonzero equilibria modulo phase are

state	(r_+, r_-)	existence	amplitude stability
traveling wave	$(\rho, 0)$ or $(0, \rho)$	$\rho^2 = -a_r/b_r > 0$	$c_r - b_r < 0$
standing wave	(ρ, ρ)	$\rho^2 = -a_r/(b_r + c_r) > 0$	$c_r - b_r > 0$

. (81)

The imaginary parts give the phase equations

$$(\theta_+)_\tau = \Im(a) + \Im(b) r_+^2 + \Im(c) r_-^2, \tag{82}$$
$$(\theta_-)_\tau = \Im(a) + \Im(b) r_-^2 + \Im(c) r_+^2, \tag{83}$$

and hence only phase drift and frequency corrections in the unforced problem. The leading physical patterns are

$$u_{\mathrm{TW}}(x,t) \sim 2\epsilon\rho \operatorname{Re}\left[\Phi e^{i(\omega_0 t \pm k_0 x + \theta)}\right], \tag{84}$$
$$u_{\mathrm{SW}}(x,t) \sim 4\epsilon\rho \cos(k_0 x + \vartheta_x) \operatorname{Re}\left[\Phi e^{i(\omega_0 t + \vartheta_t)}\right]. \tag{85}$$

6.2 Leading forcing resonances

The DC responses merely shift the linear coefficient. Define

$$\begin{aligned}\chi_0 = {} & 2\mathbf{1}_{\{n=2\}}\mathbf{1}_{\{\omega=0\}}\,\Psi^*\mathcal{Q}(\Phi, p_0^+ + p_0^-) \\ & + 2\mathbf{1}_{\{m=2\}}\mathbf{1}_{\{k=0\}}\,\Psi^*\mathcal{Q}(\Phi, s_0^+ + s_0^-).\end{aligned} \tag{86}$$

Then a is replaced by $a + \chi_0$ in both equations.

For temporal 2:1 forcing, take

$$n = 2, \qquad \omega = 2\omega_0 + \epsilon^2\delta_t, \qquad \eta_t := 2\Psi^*\mathcal{Q}\left(\overline{\Phi}, R(2\omega_0, 0)P\right). \tag{87}$$

The slow factor is $\mathrm{e}^{\mathrm{i}\delta_t\tau}$, and

$$A_\tau = A(a + b|A|^2 + c|B|^2) + \eta_t\mathrm{e}^{\mathrm{i}\delta_t\tau}\overline{B}, \tag{88a}$$
$$B_\tau = B(a + b|B|^2 + c|A|^2) + \eta_t\mathrm{e}^{\mathrm{i}\delta_t\tau}\overline{A}. \tag{88b}$$

For spatial 2:1 forcing, take

$$m = 2, \qquad k = 2k_0, \qquad \eta_x^+ = 2\Psi^*\mathcal{Q}(\Phi, R(0, 2k_0)S), \quad \eta_x^- = 2\Psi^*\mathcal{Q}(\Phi, R(0, -2k_0)\overline{S}). \tag{89}$$

Then

$$A_\tau = A(a + b|A|^2 + c|B|^2) + \eta_x^+ B, \tag{90a}$$
$$B_\tau = B(a + b|B|^2 + c|A|^2) + \eta_x^- A. \tag{90b}$$

If both 2:1 conditions hold, the contributions add:

$$A_\tau = A(a + \chi_0 + b|A|^2 + c|B|^2) + \eta_t\mathrm{e}^{\mathrm{i}\delta_t\tau}\overline{B} + \eta_x^+ B, \tag{91a}$$
$$B_\tau = B(a + \chi_0 + b|B|^2 + c|A|^2) + \eta_t\mathrm{e}^{\mathrm{i}\delta_t\tau}\overline{A} + \eta_x^- A. \tag{91b}$$

With $A = \widetilde{A}\mathrm{e}^{\mathrm{i}\delta_t\tau/2}$ and $B = \widetilde{B}\mathrm{e}^{\mathrm{i}\delta_t\tau/2}$, this becomes autonomous after the replacement $a + \chi_0 \mapsto a + \chi_0 - \mathrm{i}\delta_t/2$. Explicitly,

$$A_\tau = \mathrm{e}^{\mathrm{i}\delta_t\tau/2}\left(\widetilde{A}_\tau + \frac{\mathrm{i}\delta_t}{2}\widetilde{A}\right), \qquad \mathrm{e}^{\mathrm{i}\delta_t\tau}\overline{B} = \mathrm{e}^{\mathrm{i}\delta_t\tau/2}\overline{\widetilde{B}}, \tag{92}$$

and similarly for the second equation.

The decomposition (47) is linear, so at cubic order

$$2\mathcal{Q}(u_1, u_2^F) = 2\mathcal{Q}(u_1, u_{2,t}^F) + 2\mathcal{Q}(u_1, u_{2,x}^F). \tag{93}$$

The first temporal–spatial product occurs one order later:

$$2\mathcal{Q}(\epsilon^2 u_{2,t}^F, \epsilon^2 u_{2,x}^F) = O(\epsilon^4). \tag{94}$$

Thus (91) contains the complete leading combination. Temporal forcing couples a mode to the conjugate counter-propagating mode; spatial forcing couples the two nonconjugate traveling modes. Their phase-locked asymptotic states are the roots of the right-hand side of the autonomous form of (91); stability is determined by its 4×4 real Jacobian. No additional universal stability inequality exists without the complex values of $a, b, c, \eta_t, \eta_x^\pm$.